

1.2 Negative Numbers

Many practical situations involve numbers less than zero. Examples include:

- Temperatures below 0°C .
- Elevations below sea level.
- Financial losses or debt.
- Electrical quantities with opposite directions.

Negative numbers are written using a minus sign.

$$-1, -2, -3, -4, \dots$$

Definition

A negative number is a number less than zero.

On a number line, negative numbers lie to the left of zero, whilst positive numbers lie to the right.

$$\dots -4 \quad -3 \quad -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3 \quad 4 \dots$$

1.2.1 Comparing Negative Numbers

A common mistake is to assume that a larger digit means a larger number.

For negative numbers, the opposite is true.

$$-2 > -5$$

because -2 is closer to zero.

Worked Example

Arrange the following numbers in ascending order:

$$-1, \quad 3, \quad -7, \quad 0, \quad 5$$

Moving from left to right on the number line gives

1.2.2 Adding Negative Numbers

A useful way to think about adding positive and negative numbers is to imagine moving along a number line.

- Adding a positive number moves to the right.
- Adding a negative number moves to the left.

Worked Example

Evaluate

$$5 + (-3)$$

Start at 5 and move 3 units left.

$$5 + (-3) =$$

Worked Example

Evaluate

$$-4 + (-6)$$

Start at -4 on the number line. Adding -6 means moving 6 units further to the left.

$$-4 \rightarrow -5 \rightarrow -6 \rightarrow -7 \rightarrow -8 \rightarrow -9 \rightarrow -10$$

Therefore,

$$-4 + (-6) =$$

When both numbers are negative, we are able to use a shortcut:

Definition

When adding two negative numbers:

1. Add their magnitudes.
2. Keep the negative sign.

For example,

$$-4 + (-6) = -10$$

1.2.3 Subtracting Negative Numbers

In the previous section, adding a positive number was interpreted as moving to the right along the number line, whilst adding a negative number was interpreted as moving to the left. For example,

$$5 + 3$$

means

start at 5 and move 3 units to the right.

Similarly,

$$5 + (-3)$$

means

start at 5 and move 3 units to the left.

Subtraction can be thought of as *undoing* an addition. For example,

$$8 - 3 = 5$$

because adding 3 to 5 gives 8:

$$5 + 3 = 8$$

Subtracting 3 therefore removes the movement of 3 units to the right.

Now consider a negative number. Adding -4 means moving 4 units to the left. Therefore, subtracting -4 means undoing that leftward movement. If a leftward movement is removed, the result is a movement to the right. This is why subtracting a negative number gives the same result as adding a positive number.

Definition

Subtracting a negative changes the operation to addition.

$$a - (-b) = a + b$$

Worked Example

Evaluate

$$7 - (-4)$$

Subtracting -4 means removing a movement of 4 units to the left. Removing a leftward movement is equivalent to moving 4 units to the right. Therefore,

and so

Another example is

$$3 - (-2)$$

Subtracting -2 means removing a movement of 2 units to the left. Removing that leftward movement moves us 2 units to the right:

$$3 + 2 = 5$$

Therefore,

$$3 - (-2) = 5$$

A useful phrase to remember is

Subtracting a negative is the same as adding a positive.

1.2.4 Signs Belong to Numbers

A common mistake is to treat signs as separate symbols that can be moved independently of the numbers to which they belong.

The sign immediately to the left of a number belongs to that number. For example,

$$5 - 7 + 10$$

can be written as

$$5 + (-7) + 10$$

This shows that the expression contains the numbers

$$5, \quad -7, \quad 10$$

The signs are part of the numbers and must not be changed.

Worked Example

Evaluate

$$5 - 7 + 10$$

The signed numbers are

Working from left to right,

and

Therefore,

It would be incorrect to rewrite the expression as

$$5 - 10 + 7$$

or

$$5 - 10 - 7,$$

because this changes the signs attached to the numbers.

This idea becomes particularly important in algebra.

Worked Example

Simplify

$$3x - 5 + 2x + 7$$

The terms are

When rearranging terms, each term keeps its sign:

Combining like terms gives

When rearranging an expression, always move the sign together with the number or algebraic term.

1.2.5 Multiplying Negative Numbers

The sign of the answer depends on the number of negative signs.

Definition

$$(+) \times (+) = +$$

$$(+) \times (-) = -$$

$$(-) \times (+) = -$$

$$(-) \times (-) = +$$

Many students memorise the sign rules for multiplication without understanding why they work. One way to think about negative signs is as *reversals*.

This is similar to everyday language:

- “Move forward” = positive.
- “Reverse” = negative.
- “Reverse a reverse” = back to the original direction.

A single negative sign introduces one reversal. For example, multiplying a positive number by a negative number produces a negative result because the direction has been reversed once.

$$3 \times (-2) = -6$$

Similarly,

$$(-3) \times 2 = -6$$

In each case there is one reversal, so the answer is negative.

Now consider

$$(-3) \times (-2)$$

The first negative sign introduces a reversal. The second negative sign introduces another reversal. Two reversals cancel each other out, returning us to the original direction. Therefore,

$$(-3) \times (-2) = 6$$

Although this is only an analogy, it explains why two negative signs produce a positive result.

Worked Example

Evaluate

$$(-3) \times 4$$

There is one negative sign, so there is one reversal.

Worked Example

Evaluate

$$-5 \times -2$$

There are two negative signs, giving two reversals. The reversals cancel each other out.

1.2.6 Dividing Negative Numbers

Division follows the same ideas as multiplication. A negative sign can again be viewed as a reversal. One reversal produces a negative result. Two reversals cancel each other out and produce a positive result.

Worked Example

Evaluate

$$-20 \div 5$$

There is one negative sign, so there is one reversal.

$$-20 \div 5 =$$

Worked Example

Evaluate

$$-24 \div -6$$

There are two negative signs. The two reversals cancel each other out.

$$-24 \div -6 =$$

Notice that multiplication and division follow exactly the same reasoning. Whenever two negative signs appear, the two reversals cancel each other out.

1.2.7 Negative Numbers and BIDMAS

Negative numbers do not change the order of operations.

Worked Example

Evaluate

$$-3 + 2 \times 5$$

Perform the multiplication first:

Then add:

Therefore,

Worked Example

Evaluate

$$(-3)^2$$

The brackets indicate that the entire number is squared.

Notice the difference between

$$(-3)^2 = 9$$

and

$$-3^2 = -9$$

since the index is applied before the leading negative sign.

1.2.8 Common Mistakes

Common errors involving negative numbers include:

- Assuming -5 is larger than -2 .
- Forgetting that subtracting a negative becomes addition.
- Moving numbers without moving their signs.

- Assuming a negative multiplied by a negative is negative.
- Ignoring brackets when squaring negative numbers.
- Applying BIDMAS incorrectly when negative signs are present.

1.2.9 Engineering Application

Engineering Application

Negative numbers frequently occur in engineering calculations.

For example, a temperature sensor may record a temperature of

$$-8^{\circ}\text{C}$$

overnight and

$$4^{\circ}\text{C}$$

the following afternoon.

The temperature increase is

$$4 - (-8) = 12^{\circ}\text{C}$$

In electrical engineering, voltages and currents may be assigned positive or negative values depending on the chosen reference direction.

Correct handling of negative numbers is therefore essential for obtaining meaningful engineering results.

1.2.10 Exercises

Exercise

Evaluate the following expressions.

1. $5 + (-8)$
2. $-3 + 7$
3. $-4 + (-6)$
4. $9 - (-2)$
5. $-12 - 5$
6. $(-3) \times 4$
7. $(-5) \times (-6)$
8. $-24 \div 6$
9. $-21 \div (-7)$
10. $(-2)^3$
11. $(-4)^2$
12. -4^2
13. $-5 + 3 \times 4$
14. $18 + (-8) \div 2$
15. $7 - (-3) \times 2$
16. $5 - 7 + 10$
17. $8 - 12 + 5$
18. $-4 + 9 - 7$
19. $-3 - 5 + 8$
20. $12 - 15 + 4 - 3$
21. Simplify $3x - 5 + 2x + 7$
22. Simplify $4a + 8 - 2a - 3$
23. Simplify $7m - 4 + 3m - 6$
24. Simplify $5y - 8 - y + 11$
25. Simplify $9p - 12 - 4p + 7$

Answers to Exercise 1.2.10

1. -3
2. 4
3. -10
4. 11
5. -17
6. -12
7. 30
8. -4
9. 3
10. -8
11. 16
12. -16
13. 7
14. 14
15. 13
16. 8
17. 1
18. -2
19. 0
20. -2
21. $5x + 2$
22. $2a + 5$
23. $10m - 10$
24. $4y + 3$
25. $5p - 5$